MILITARY SPECIFICATION

BORON MONOFILAMENT, CONTINUOUS, VAPOR DEPOSITED

1. SCOPE

1.1 This specification covers one type of high quality boron filament 0.0040 inch in nominal diameter produced by the hydrogen reduction of borontrichloride on a tungsten substrate.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS
Federal

Federal Test Method Standard No. 406	Plastics: Methods of Testing

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Mechanical properties. The boron filament material supplied shall have a nominal diameter of 0.0040 inch with a tolerance of ±0.0002 inch. The lot average tensile strength shall be 450,000 psi, minimum, with no individual value below 350,000 psi. Breaking load values in pounds may be used for acceptance if they meet the requirements of table I. The filament shall meet the other requirements of table I.

3.2 Splices. Unless otherwise specified, splicing of boron filaments shall be permitted and shall comply with the requirements of type (3.2.1), compatibility (3.2.2), heat (3.2.3), conformability (3.2.4) and frequency (3.2.5) to achieve proper package lengths.

3.2.1 Type. 1 inch to 2 inch overlap splice with uniform resin (adhesive) application free of agglomerates. Splices shall be capable of being drawn through a 0.015 inch wide slot.

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Table 1. Boron filament properties.

Property	Requirement	
Breaking Load, lbs. Filament Diameter, inch	Nominal Lot Average 1/	Minimum Acceptable 2/
0.0038	5.12	3.98
0.0039	5.37	4.18
0.0040	5.66	4.40
0.0041	5.93	4.62
0.0042	6.22	4.84
Variation, max acceptable, %	20	
Modulus of elasticity, tension, psi	58×10^6	55×10^6
Diameter, inch		
Nominal	0.0040	
Minimum	0.0038	
Maximum	0.0042	
Twist, max acceptable, turns/10 ft.	1.0	
Density, max lbs/inch^3	0.098	

1/ Equating to a lot average tensile strength of 450,000 psi for nominal 0.0040 inch diameter filament (0.0040 ± 0.0002).

2/ Based on the actual diameter with the lowest acceptable tensile strength, 350,000 psi.

3.2.2 Compatibility. Splicing material shall be a polyimide resin based material cured sufficiently to be resistant to solvents used in polyimide resin systems and to other resin systems and their solvents.

3.2.3 Heat requirement. Splices shall be capable of withstanding a minimum of a 64 gram load at 400°F for not less than 10 minutes and must not emit volatiles at temperatures up to 375°F.

3.2.4 Conformability. The quality of splices shall be such that breaks or splits will not occur after being wound on the shipping spool by the filament producer. The producer shall perform inspections as frequently as necessary to assure conformance with the requirements of this specification.

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3.2.5 Frequency. Average distance between splices shall be at least 3000 ft with a minimum length of 1000 ft.

3.3 Workmanship. The material shall be uniform and free from imperfections and defects with adversely affect performance and use in subsequent processing. The tungsten substrate and the reacting gases shall be of sufficient quality to ensure a product which will meet the requirements of this specification.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the procuring activity. The procuring activity reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure materials and services conform to the prescribed requirements.

4.1.1 Inspections records shall be kept complete and available to the procuring agency at all times.

4.2 Lot. A lot shall consist of a quantity of boron filament collected for shipment to a single use or material processor.

4.2.1 Filament production run. A filament run consists of all filament manufactured in continuous operation of a single reactor.

4.3 Acceptance test. Tests on each order or filament production run of material consist of, but need not be limited to, the following examinations and tests:

a. Filament tensile test (4.4.1)
b. Filament variation (4.4.2)
c. Filament tensile modulus test (4.4.3)
d. Filament diameter measurement (4.4.4)
e. Density measurements (4.4.5)
f. Filament twist test (4.4.6)

4.4 Property testing.

4.4.1 Tensile test. Tensile strength shall be measured for each continuous filament production machine run (see 4.2.1). Tests shall be conducted using

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a specimen with a one-inch gage length and tested at a loading rate of 0.05 to 0.50 inches per minute. A average tensile load, $\bar{X}_{10}$, shall be determined from 10-individual tensile specimens taken from each production machine run. The standard deviation, σ, shall also be calculated for each sample set as follows.

$$\sigma = \text{positive square root of } \frac{1}{n-1} \sum_{1}^{n} (x - \bar{X}_{10})^2$$

where n is the number of samples.

4.4.2 Variation. Coefficient of variation shall be computed according to the following formula. It shall be computed for each order of shipment made, which ever shall be the smaller.

$$\% \text{ average coefficient of variation} = \frac{1}{L} \sum_{1}^{L} \frac{\sigma}{\bar{X}_{10}} \times 100$$

where L is the number of production machine runs within a specific shipment.

4.4.3 Modulus of elasticity, tension. The tension modulus of elasticity of the boron filaments shall be based either on sonic measurements or specimens using long gage lengths (10 inches or greater) from which the stress-strain values can be obtained. The frequency of modulus determinations shall be established as that amount necessary to assure conformance to the requirements of this specification. In case of dispute, stress-strain measurements using long gage lengths shall be used as the referee method.

4.4.4 Diameter. The diameter shall be measured optically or by other suitable means to assure adherence to both the tolerances required in the specification and to provide the base to calculate that the minimum tensile strength requirements are met.

4.4.5 Density. Tests shall be performed in accordance with Federal Test Method Standard No. 406, Method 5011 or 5012, or equivalent. Frequency of testing shall be established to assure conformance with the requirements of this specification.

4.4.6 Twist. After removing, under tension, and discarding approximately 30 ft of filament from the free end of the spool, tag the free end and unwind, without permitting to untwist, a section of filament 10 ft long. Secure the 10 ft length in a vertical position, in the absence of drafts; free the lower end and count the number of revolutions made about the vertical axis by the free, lower end. Frequency of testing shall be established to assure compliance with this requirement of the specification.

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4.5 Conformance certification data. Test reports and certificates of conformance furnished in accordance with these requirements shall include the following identifying information:

a. Purchase order number
b. Materials specification number and revision
c. Manufacturer's spool number
d. Average test values for each spool.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be accomplished by the manufacturer to the best commercial methods to assure transportability and ease of handling which maintaining material shipped in its most usable form.

5.2 Filament spool.

5.2.1 Spool diameter. The boron filament shall be supplied on spools of a diameter not less than 8 inches, Hubbard IM20 or equivalent.

5.2.2 Winding tension. Boron filaments shall be wound under uniform tension on the shipping spools to insure a transportable, handleable and usable material.

5.2.3 Winding pattern. Filament shall be closely wound in layers on the spools but adjacent turns within a layer need not necessarily be touching; shall be wound so as to avoid producing kinks, waves and sharp bends; and shall be free to unwind without restriction caused by overlapping or wedging.

5.2.4 Liner interleave. Filament may be wound with a paper liner interleave inserted at the end of each layer. Liner material shall be of such stock as to impart no contaminates to the filament or filament surface with would affect composite properties.

5.2.5 Filament length per spool. Unless otherwise specified, each spool shall contain 35,000 ft. ±10 percent of boron filament. The number of splices per spool permitted to achieve the desired length of 35,000 feet shall not exceed 10 and shall be recorded by the manufacturer and made available to the purchaser upon request.

5.2.6 Protective packaging. Unless otherwise specified, a protective foamed plastic cover shall be placed over the open face of the material to protect it from damage during shipping. Shipment packages may contain

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more than one spool, and shall be packed in accordance with the contractor's best commercial practices.

5.3 Marking.

5.3.1 Marking of spools. Each spool shall be marked with the following information:

BORON MONOFILAMENT, CONTINUOUS, VAPOR DEPOSITED
Specification Number
Spool Number
Average unweighted tensile strength
Diameter
Length
Manufacturer's name, trademark, or symbol

5.3.2 Marking of containers. Each shipping container shall be marked with the following information:

BORON MONOFILAMENT CONTINUOUS, VAPOR DEPOSITED
Purchase Order Number
Specification Number
Manufacturer's Identification Number
Manufacturer's name, trademark, or symbol

FRAGILE - HANDLE WITH CARE

6. NOTES

6.1 Intended use. The material procured in accordance with this specification is intended for use in airframe and aerospace primary structural components and in similar assemblies where high stiffness and high strength-to-weight ratios are required.

6.2 Ordering data. Procurement documents shall specify the title, number and date of this specification.

Custodian:
Air Force - 11

Reviewer:
Air Force - 84

Preparing activity:
Air Force - 11

Project Number 9330-F486

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC-WPAFB-OCT 67 2M

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